H8: Conservation laws

8.1 charge and energy

8.1.1 the continuity equation

continuity equation

The charge in a volume V is: $Q(t) = \int_{\mathcal{V}} \rho(\mathbf{r}, t)\, d\tau,$

and conservation of charge says: $\frac{dQ}{dt} = -\oint_{\mathcal{S}} \mathbf{J} \cdot d\mathbf{a}.$

thus we find: $\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}.$

8.1.2 Poynting's theorem

Energy in E and B

The work needed to assemble a static charge distribution is given by:

$$W_e = \frac{\epsilon_0}{2} \int E^2\, d\tau,$$

And for the magnetic field:

$$W_m = \frac{1}{2\mu_0} \int B^2\, d\tau,$$

this suggests the total energy stored, per unit volume is given by:

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right)$$

Poynting's theorem

If a configuration of charges changes a bit over a time dt, then we have:

$$dW = \mathbf{F} \cdot d\mathbf{l} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v}\, dt = q\mathbf{E} \cdot \mathbf{v}\, dt.$$

Now we take $q \to \rho d\tau$ and $\rho v \to J$, so the work done on charges in a volume V is:

$$\frac{dW}{dt} = \int_{\mathcal{V}} (\mathbf{E} \cdot \mathbf{J})\, d\tau.$$

Now, via Ampère-Maxwell law:

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}.$$

From product rule 6,

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}).$$

Invoking Faraday's law ($\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$), it follows that

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = -\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B}).$$

Meanwhile,

$$\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t}(B^2), \quad \text{and} \quad \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t}(E^2),$$

so

$$\mathbf{E} \cdot \mathbf{J} = -\frac{1}{2} \frac{\partial}{\partial t}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) - \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}).$$

This implies that:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) d\tau - \frac{1}{\mu_0} \oint_{\mathcal{S}} (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a},$$

ie: the work done on the charges by electromagnetic force is equal to the decrease in energy remaining in the fields, less the energy that flowed out through the surface

Poynting vector

Define S the energy flux density: $\mathbf{S} \equiv \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}).$

then Poynting's theorem becomes:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} u\, d\tau - \oint_{\mathcal{S}} \mathbf{S} \cdot d\mathbf{a}.$$

continuity equation for energy

if there is no work done, thus dW/dt=0, we find that: $\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}.$

8.2 momentum	
8.2.1 newton's third law in electrodynamics	
newtons third law in em	consider two moving charges q_1 and q_2 in perpendicular paths > the magnetic fields of the charges exert a force on each other > due to right-hand rule, one is pointed upwards and other to the right > the forces are equal, but NOT opposite to one another > violates the third law of Newton
8.2.2 Maxwell's stress tensor	
Maxwell's stress tensor	the total electromagnetic force on changes in volume V is: $$\mathbf{F} = \int_{\mathcal{V}} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \rho \, d\tau = \int_{\mathcal{V}} (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) \, d\tau.$$ the force per unit volume is: $$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}.$$ express this in terms of fields, and eliminate ρ and J using maxwells 1st and 4th eq: $$\mathbf{f} = \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \times \mathbf{B}.$$ now: $$\frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) = \left(\frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B} \right) + \left(\mathbf{E} \times \frac{\partial \mathbf{B}}{\partial t} \right)$$ and Faraday's first law says: $$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E},$$ so: $$\frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B} = \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) + \mathbf{E} \times (\nabla \times \mathbf{E}).$$ thus: $$\mathbf{f} = \epsilon_0 \left[(\nabla \cdot \mathbf{E}) \mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E}) \right] - \frac{1}{\mu_0} \left[\mathbf{B} \times (\nabla \times \mathbf{B}) \right] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}).$$ furthermore we know: $$\nabla (E^2) = 2 (\mathbf{E} \cdot \nabla) \mathbf{E} + 2 \mathbf{E} \times (\nabla \times \mathbf{E})$$ so: $$\mathbf{E} \times (\nabla \times \mathbf{E}) = \frac{1}{2} \nabla (E^2) - (\mathbf{E} \cdot \nabla) \mathbf{E},$$ giving us: $$\mathbf{f} = \epsilon_0 \left[(\nabla \cdot \mathbf{E}) \mathbf{E} + (\mathbf{E} \cdot \nabla) \mathbf{E} \right] + \frac{1}{\mu_0} \left[(\nabla \cdot \mathbf{B}) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{B} \right]$$ $$- \frac{1}{2} \nabla \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}).$$ define Maxwell's stress tensor: $$T_{ij} \equiv \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right)$$ which has the structure: $$T_{xx} = \frac{1}{2} \epsilon_0 \left(E_x^2 - E_y^2 - E_z^2 \right) + \frac{1}{2\mu_0} \left(B_x^2 - B_y^2 - B_z^2 \right)$$ $$T_{xy} = \epsilon_0 (E_x E_y) + \frac{1}{\mu_0} (B_x B_y),$$ the divergence of this tensor is: $$\left(\nabla \cdot \overleftrightarrow{\mathbf{T}} \right)_j = \epsilon_0 \left[(\nabla \cdot \mathbf{E}) E_j + (\mathbf{E} \cdot \nabla) E_j - \frac{1}{2} \nabla_j E^2 \right] + \frac{1}{\mu_0} \left[(\nabla \cdot \mathbf{B}) B_j + (\mathbf{B} \cdot \nabla) B_j - \frac{1}{2} \nabla_j B^2 \right]$$ therefore we can simplify: $$\mathbf{f} = \nabla \cdot \overleftrightarrow{\mathbf{T}} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}$$ the total electromagnetic force on charges in volume V is then: $$\mathbf{F} = \oint_{\mathcal{S}} \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_{\mathcal{V}} \mathbf{S} \, d\tau.$$
